

Solutions – Quiz 1

ECON 320 – Introduction to Mathematical Economics

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Q1 Absolute-Value Inequality

Problem. Solve the inequality

$$|2x - 5| < 7$$

and represent the solution set on the real line. Interpret in economic terms if x is quantity and the inequality represents a feasible demand constraint.

Solution

Recall that for any real u and any $a > 0$, $|u| < a \iff -a < u < a$. Let $u = 2x - 5$ and $a = 7$:

$$-7 < 2x - 5 < 7.$$

Add 5 to all three parts:

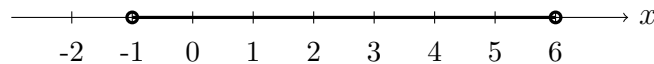
$$-2 < 2x < 12.$$

Divide by 2:

$$-1 < x < 6.$$

Hence, the solution set is the open interval $(-1, 6)$.

Number-line representation. (Open endpoints at $x = -1$ and $x = 6$)



Economic interpretation. If x denotes quantity and the inequality encodes a feasible *demand* region, then all quantities $x \in (-1, 6)$ are feasible. Since negative quantities are typically not economically meaningful, one would usually intersect with $x \geq 0$, yielding $x \in [0, 6)$ in practical applications.

Q2. Feasible Set and Resource Allocation

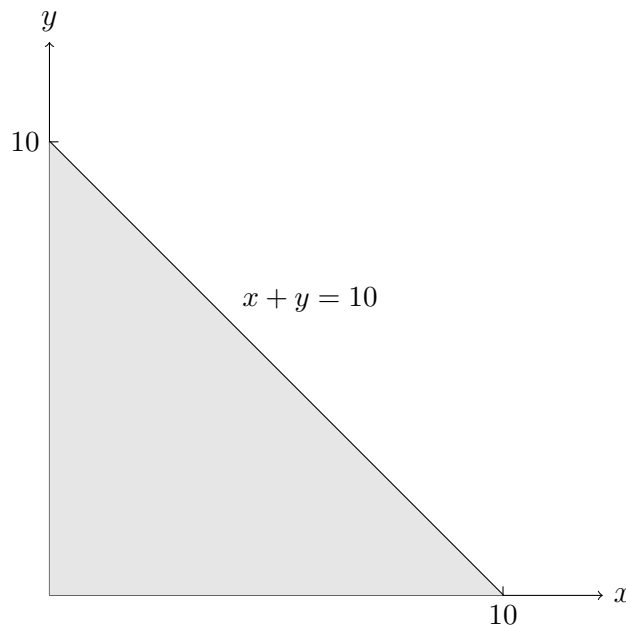
Problem. Consider

$$S = \{(x, y) \in \mathbb{R}^2 \mid x + y \leq 10, x \geq 0, y \geq 0\}.$$

- (a) Sketch S .
- (b) If $y = 2$, what values of x are admissible?
- (c) Interpret this as a resource allocation problem.

Solution

- (a) The boundary $x + y = 10$ has intercepts $(10, 0)$ and $(0, 10)$. Together with $x \geq 0$ and $y \geq 0$, the feasible set is the filled right triangle in the first quadrant below (and including) the line $x + y = 10$.



- (b) If $y = 2$, the constraint $x + y \leq 10$ becomes $x + 2 \leq 10$, i.e. $x \leq 8$. With $x \geq 0$, the admissible set is

$$0 \leq x \leq 8.$$

- (c) **Resource allocation interpretation.** Suppose a planner has 10 units of a resource to allocate between two nonnegative uses x and y . The set S is the budget (or capacity) set. Any point below the line uses fewer than 10 units; points on the line use exactly 10 units.

Q3. Implicit Demand and Total Differentiation

Problem. Consider the demand relation

$$Q^d(P, Y) = Y - P^2 = 0,$$

where P is price and Y is income. (a) Derive $\frac{dP}{dY}$. (b) What does its sign imply about the relationship between income and the equilibrium price?

Solution

From $Y - P^2 = 0$ we can (for $P \geq 0$) write $P = \sqrt{Y}$, which already suggests P increases with Y for $Y \geq 0$. Using *implicit differentiation*:

$$\frac{d}{dY}(Y - P^2) = 1 - 2P \frac{dP}{dY} = 0 \implies \frac{dP}{dY} = \frac{1}{2P}.$$

Equivalently, using $P = \sqrt{Y}$,

$$\frac{dP}{dY} = \frac{1}{2\sqrt{Y}}, \quad (Y > 0).$$

Sign and interpretation. Since $P > 0$ at interior solutions, $\frac{dP}{dY} > 0$: as income rises, the equilibrium price consistent with $Y - P^2 = 0$ increases. Moreover, $\frac{dP}{dY}$ is *diminishing* in P (and in Y): the marginal effect of income on price is larger at lower income levels and shrinks as income grows.